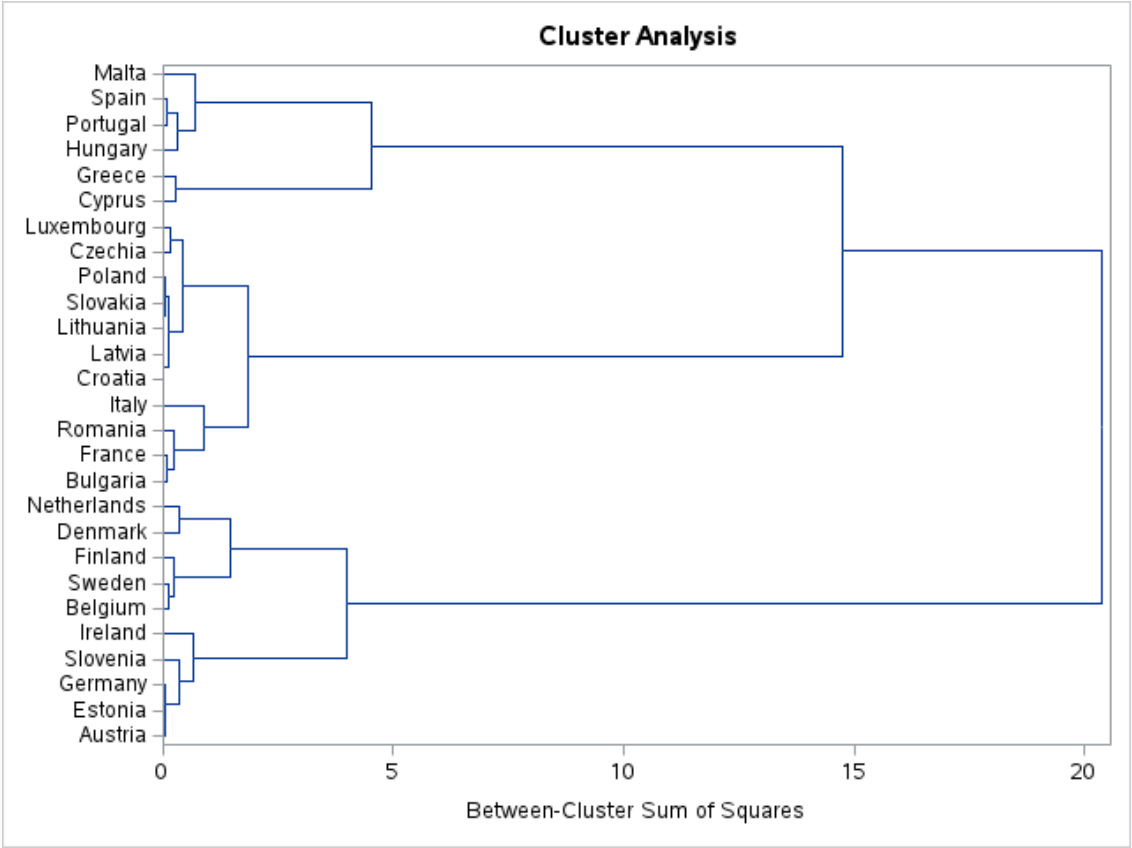


Hierarchical Cluster Analysis

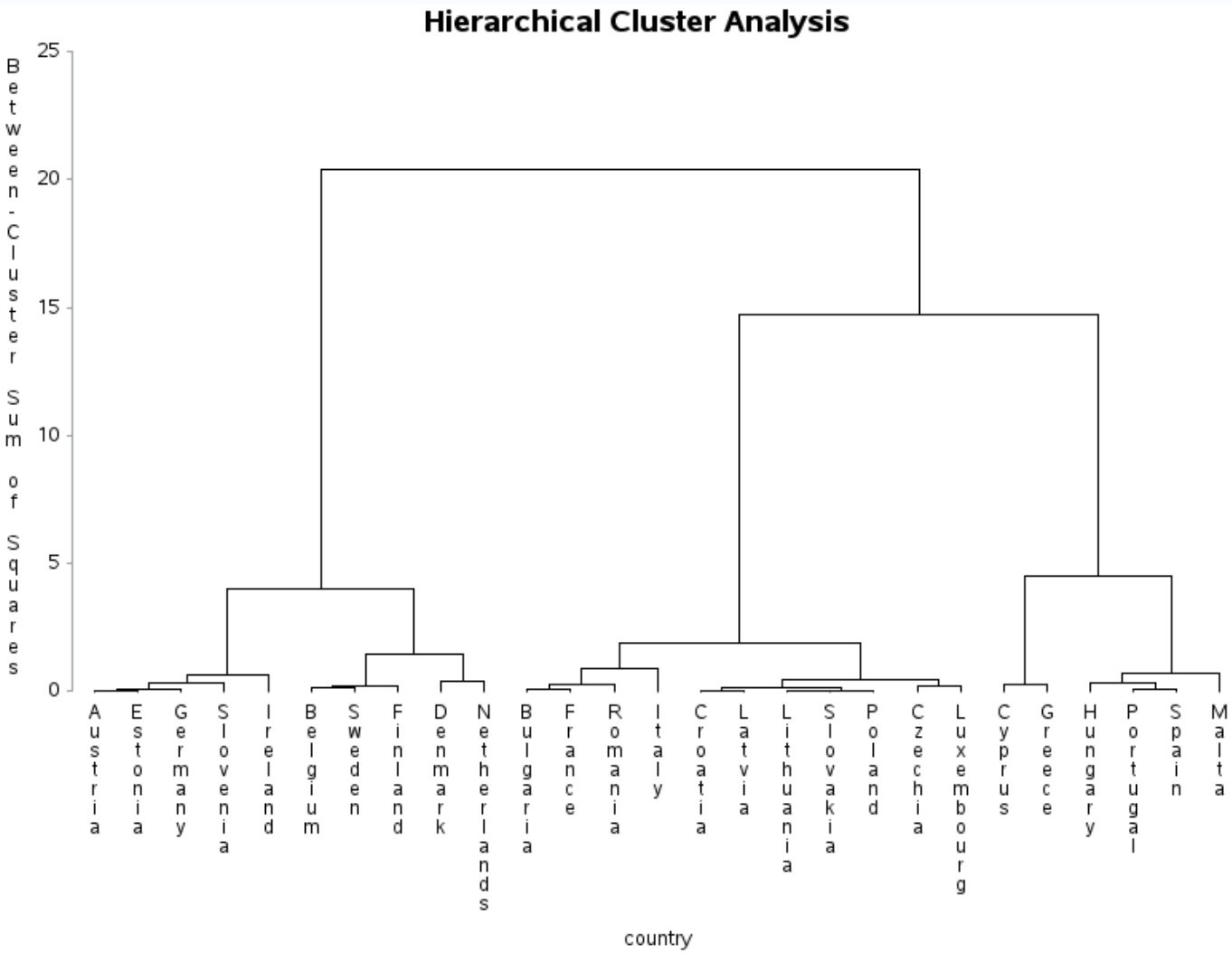
Variable	Mean	Standard Deviation	Skewness	Kurtosis	Bimodality
Factor1	0	1.0000	-0.00653	-0.4859	0.3456
Factor2	0	1.0000	0.4112	0.1213	0.3339

Root-Mean-Square Total-Sample Standard Deviation	1
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Cluster History							
Number of Clusters	Clusters Joined		Freq	New Cluster RMS Std Dev	Semipartial R-Square	R-Square	Between Cluster Sum of Squares
26	Croatia	Latvia	2	0.0615	0.0001	1.00	0.0076
25	Lithuania	Slovakia	2	0.0761	0.0002	1.00	0.0116
24	CL25	Poland	3	0.0968	0.0005	.999	0.0259
23	Austria	Estonia	2	0.1150	0.0005	.999	0.0265
22	CL23	Germany	3	0.1489	0.0012	.997	0.0622
21	Portugal	Spain	2	0.1818	0.0013	.996	0.0661
20	Bulgaria	France	2	0.1877	0.0014	.995	0.0705
19	CL26	CL24	5	0.1377	0.0021	.993	0.1067
18	Belgium	Sweden	2	0.2364	0.0022	.991	0.1118
17	Czechia	Luxembourg	2	0.2955	0.0034	.987	0.1746
16	CL18	Finland	3	0.2885	0.0043	.983	0.221
15	CL20	Romania	3	0.2788	0.0046	.978	0.2405
14	Cyprus	Greece	2	0.3763	0.0054	.973	0.2831
13	Hungary	CL21	3	0.3121	0.0062	.967	0.3235
12	CL22	Slovenia	4	0.2673	0.0065	.960	0.34
11	Denmark	Netherlands	2	0.4248	0.0069	.953	0.3609
10	CL19	CL17	7	0.2531	0.0085	.945	0.4422
9	CL12	Ireland	5	0.3649	0.0122	.932	0.6367
8	CL13	Malta	4	0.4228	0.0131	.919	0.683
7	CL15	Italy	4	0.4482	0.0172	.902	0.8942
6	CL16	CL11	5	0.5178	0.0279	.874	1.4509
5	CL7	CL10	11	0.4369	0.0355	.839	1.8441
4	CL9	CL6	10	0.6325	0.0767	.762	3.9902
3	CL14	CL8	6	0.7663	0.0869	.675	4.517
2	CL5	CL3	17	0.8737	0.2834	.392	14.737
1	CL4	CL2	27	1.0000	0.3918	.000	20.373



Ward's Minimum Variance Cluster Analysis



Hierarchical Cluster Analysis

CLUSTER=1

Obs	country	Factor1	Factor2
1	Croatia	-0.53487	-0.43408
2	Latvia	-0.41372	-0.41269
3	Lithuania	-0.19602	-0.54284
4	Slovakia	-0.34756	-0.55728
5	Poland	-0.17513	-0.72165
6	Bulgaria	-0.89383	-0.81781
7	France	-0.51847	-0.81917
8	Czechia	0.18050	-0.41975
9	Luxembourg	0.20457	-1.01016
10	Romania	-0.67870	-1.41848
11	Italy	-0.55691	-2.10139

CLUSTER=2

Obs	country	Factor1	Factor2
12	Austria	0.92695	-0.26146
13	Estonia	0.70077	-0.21968
14	Germany	0.58580	-0.44380
15	Slovenia	1.39956	-0.43289
16	Ireland	0.32278	0.33794

CLUSTER=3

Obs	country	Factor1	Factor2
17	Portugal	-0.74687	-0.00970
18	Spain	-0.45027	0.20056
19	Hungary	-1.14054	0.53312
20	Malta	-0.61841	1.18194

CLUSTER=4

Obs	country	Factor1	Factor2
21	Belgium	1.58932	0.23557
22	Sweden	1.16398	0.44225
23	Finland	1.69573	0.81819
24	Denmark	1.48062	1.85440
25	Netherlands	0.87728	1.25632

CLUSTER=5

Obs	country	Factor1	Factor2
26	Cyprus	-2.13442	2.19603
27	Greece	-1.72213	1.56651

Hierarchical Cluster Analysis

CLUSTER=1

Variable	Mean
Factor1	-0.3572853
Factor2	-0.8413910

CLUSTER=2

Variable	Mean
Factor1	0.7871725
Factor2	-0.2039790

CLUSTER=3

Variable	Mean
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Variable	Mean
Factor1	-0.7390238
Factor2	0.4764806

CLUSTER=4

Variable	Mean
Factor1	1.3613852
Factor2	0.9213470

CLUSTER=5

Variable	Mean
Factor1	-1.9282775
Factor2	1.8812691

Kmeans - Initial Seeds from &Algorithm

Replace=NONE Radius=0 Maxclusters=5 Maxiter=50 Converge=0.02

Initial Seeds		
Cluster	Factor1	Factor2
1	-0.357285292	-0.841390996
2	0.787172530	-0.203978951
3	-0.739023850	0.476480647
4	1.361385186	0.921346993
5	-1.928277484	1.881269075

Minimum Distance Between Initial Seeds =	1.26336
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Iteration History						
Iteration	Criterion	Relative Change in Cluster Seeds				
		1	2	3	4	5
1	0.3928	0.0541	0.0849	0	0	0
2	0.3900	0	0	0	0	0

Convergence criterion is satisfied.

Criterion Based on Final Seeds =	0.3900
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Cluster Summary						
Cluster	Frequency	RMS Std Deviation	Maximum Distance from Seed to Observation	Radius Exceeded	Nearest Cluster	Distance Between Cluster Centroids
1	10	0.4284	1.2265		2	1.2720
2	6	0.3756	0.7391		1	1.2720
3	4	0.4228	0.7157		1	1.3990
4	5	0.5178	0.9406		2	1.3434
5	2	0.3763	0.3763		3	1.8406

Statistics for Variables				
Variable	Total STD	Within STD	R-Square	RSQ/(1-RSQ)
Factor1	1.00000	0.34366	0.900069	9.006874
Factor2	1.00000	0.50529	0.783962	3.628818
OVER-ALL	1.00000	0.43210	0.842015	5.329731

Pseudo F Statistic =	29.31
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Approximate Expected Over-All R-Squared =	0.85920
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Cubic Clustering Criterion =	-0.717
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WARNING: The two values above are invalid for correlated variables.

Cluster Means		
Cluster	Factor1	Factor2
1	-0.411063660	-0.883555548
2	0.686060172	-0.239940038
3	-0.739023850	0.476480647
4	1.361385186	0.921346993
5	-1.928277484	1.881269075

Cluster Standard Deviations		
Cluster	Factor1	Factor2
1	0.3051016650	0.5234767404
2	0.4393056646	0.2986410466
3	0.2939410956	0.5206985463
4	0.3358894271	0.6506297793
5	0.2915348675	0.4451342275

Kmeans - Initial Seeds from &Algorithm

Cluster=1

Obs	country	Factor1	Factor2
1	Bulgaria	-0.89383	-0.81781
2	Croatia	-0.53487	-0.43408
3	France	-0.51847	-0.81917
4	Italy	-0.55691	-2.10139
5	Latvia	-0.41372	-0.41269
6	Lithuania	-0.19602	-0.54284
7	Luxembourg	0.20457	-1.01016
8	Poland	-0.17513	-0.72165
9	Romania	-0.67870	-1.41848
10	Slovakia	-0.34756	-0.55728

Cluster=2

Obs	country	Factor1	Factor2
11	Austria	0.92695	-0.26146
12	Czechia	0.18050	-0.41975
13	Estonia	0.70077	-0.21968
14	Germany	0.58580	-0.44380
15	Ireland	0.32278	0.33794
16	Slovenia	1.39956	-0.43289

Cluster=3

Obs	country	Factor1	Factor2
17	Hungary	-1.14054	0.53312
18	Malta	-0.61841	1.18194
19	Portugal	-0.74687	-0.00970
20	Spain	-0.45027	0.20056

Cluster=4

Obs	country	Factor1	Factor2
21	Belgium	1.58932	0.23557
22	Denmark	1.48062	1.85440
23	Finland	1.69573	0.81819
24	Netherlands	0.87728	1.25632
25	Sweden	1.16398	0.44225

Cluster=5

Obs	country	Factor1	Factor2
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Obs	country	Factor1	Factor2
26	Cyprus	-2.13442	2.19603
27	Greece	-1.72213	1.56651

Kmeans - Initial Seeds from &Algorithm

Cluster=1

Variable	N	Mean	Median	Mode	Minimum	Maximum	Std Dev	Variance	10th Pctl	90th Pctl
Factor1	10	-0.4110637	-0.4660936	.	-0.8938345	0.2045706	0.3051017	0.0930870	-0.7862676	0.0147184
Factor2	10	-0.8835555	-0.7697316	.	-2.1013912	-0.4126916	0.5234767	0.2740279	-1.7599338	-0.4233873

Cluster=2

Variable	N	Mean	Median	Mode	Minimum	Maximum	Std Dev	Variance	10th Pctl	90th Pctl
Factor1	6	0.6860602	0.6432845	.	0.1804984	1.3995610	0.4393057	0.1929895	0.1804984	1.3995610
Factor2	6	-0.2399400	-0.3406042	.	-0.4437957	0.3379370	0.2986410	0.0891865	-0.4437957	0.3379370

Cluster=3

Variable	N	Mean	Median	Mode	Minimum	Maximum	Std Dev	Variance	10th Pctl	90th Pctl
Factor1	4	-0.7390238	-0.6826387	.	-1.1405438	-0.4502743	0.2939411	0.0864014	-1.1405438	-0.4502743
Factor2	4	0.4764806	0.3668427	.	-0.0097000	1.1819372	0.5206985	0.2711270	-0.0097000	1.1819372

Cluster=4

Variable	N	Mean	Median	Mode	Minimum	Maximum	Std Dev	Variance	10th Pctl	90th Pctl
Factor1	5	1.3613852	1.4806186	.	0.8772763	1.6957313	0.3358894	0.1128217	0.8772763	1.6957313
Factor2	5	0.9213470	0.8181945	.	0.2355657	1.8544029	0.6506298	0.4233191	0.2355657	1.8544029

Cluster=5

Variable	N	Mean	Median	Mode	Minimum	Maximum	Std Dev	Variance	10th Pctl	90th Pctl
Factor1	2	-1.9282775	-1.9282775	.	-2.1344238	-1.7221312	0.2915349	0.0849926	-2.1344238	-1.7221312
Factor2	2	1.8812691	1.8812691	.	1.5665116	2.1960265	0.4451342	0.1981445	1.5665116	2.1960265